

Module 1

SAP-1 Architecture

The SAP-1 design contains the basic necessities for a functional microprocessor. In this laboratory activity, a Simple-As-Possible (SAP)-1 computer will be designed in Verilog HDL.

Learning objectives:

- To design an 8-bit computer that can perform simple arithmetic operations using the SAP-1 architecture
- To design the processing unit, control unit, memory to store data and instructions, and a central bus
- To perform a simple algorithm such as instruction execution cycle to process the instruction of a program

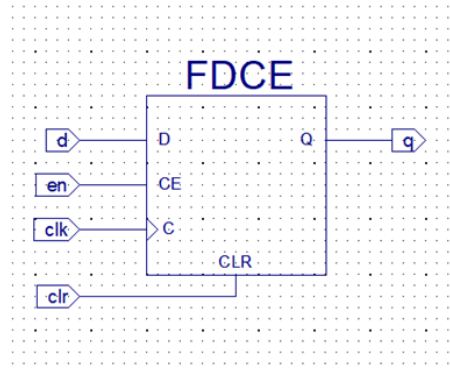
1. Registers

To be able to store a byte (8-bits) of data, a combination of eight flip-flops is needed to work as a unit. A register is a circuit with two or more D flip-flops connected together to work exactly the same way it holds data and can be synchronized by the same clock and enable signals. Each flip-flops that is grouped together to form a register can hold different bit and is synchronized by the same clock cycle.

In this activity an 8-bit register with parallel load and asynchronous clear will be designed using Verilog HDL.

Exercises

Exercises 1: Start using XILINX ISE, create new (Schematic) project and place D flip-flop with clear and enable pins (FDCE). Verify the behavior of D flip-flop using the given test-bench.

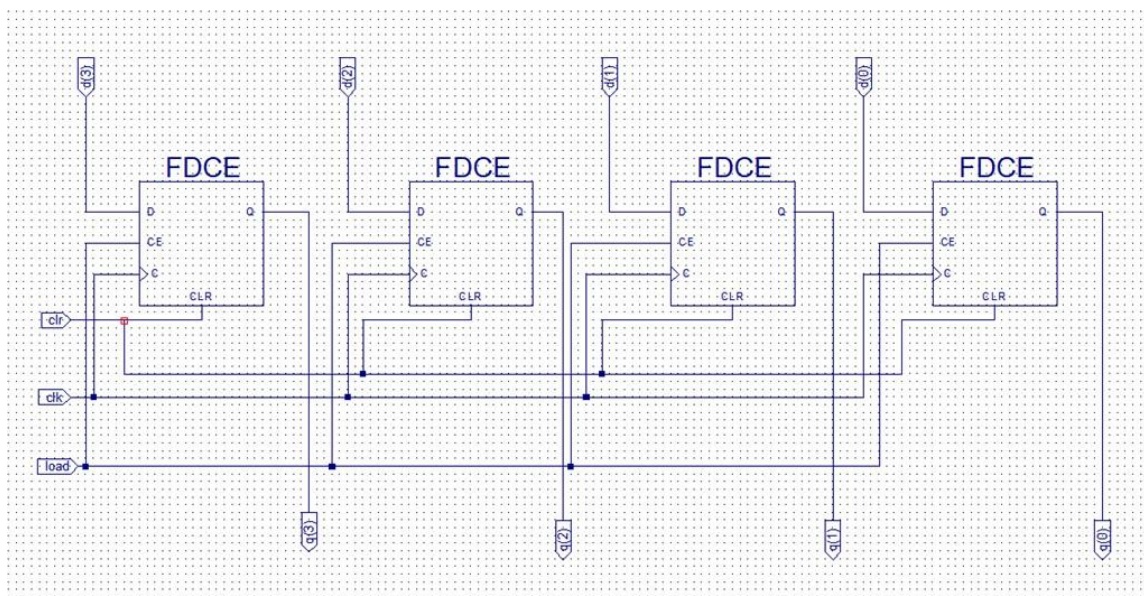


Circuit Schematic

This design element is a single D-type flip-flop with clock enable and asynchronous clear. When clock enable (CE) is High and asynchronous clear (CLR) is Low, the data on the data input (D) of this design element is transferred to the corresponding data output (Q) during the Low-to-High clock (C) transition. When CLR is High, it overrides all other inputs and resets the data output (Q) Low. When CE is Low, clock transitions are ignored.

Inputs				Output
CLR	CE	D	CLK	Q
1	X	X	X	0
0	0	X	X	NC
0	1	D	HIGH	D

Exercises 2: Design of a 4-bit register with parallel load and asynchronous clear circuit.



clr	load	Operation
1	X	Reset register to zero asynchronously
0	0	No Change
0	1	Load in a value at rising edge of clock

Design Constraints

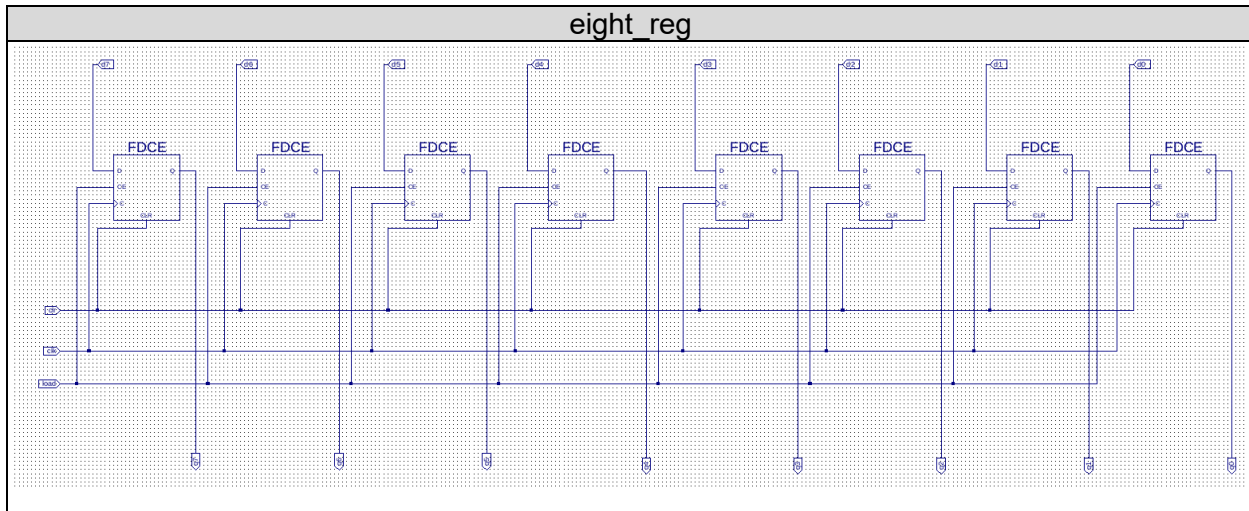
- Design of an 8-bit register with parallel load and asynchronous clear

Constraints 1: <identify some valid constraints that may affect the design if any>

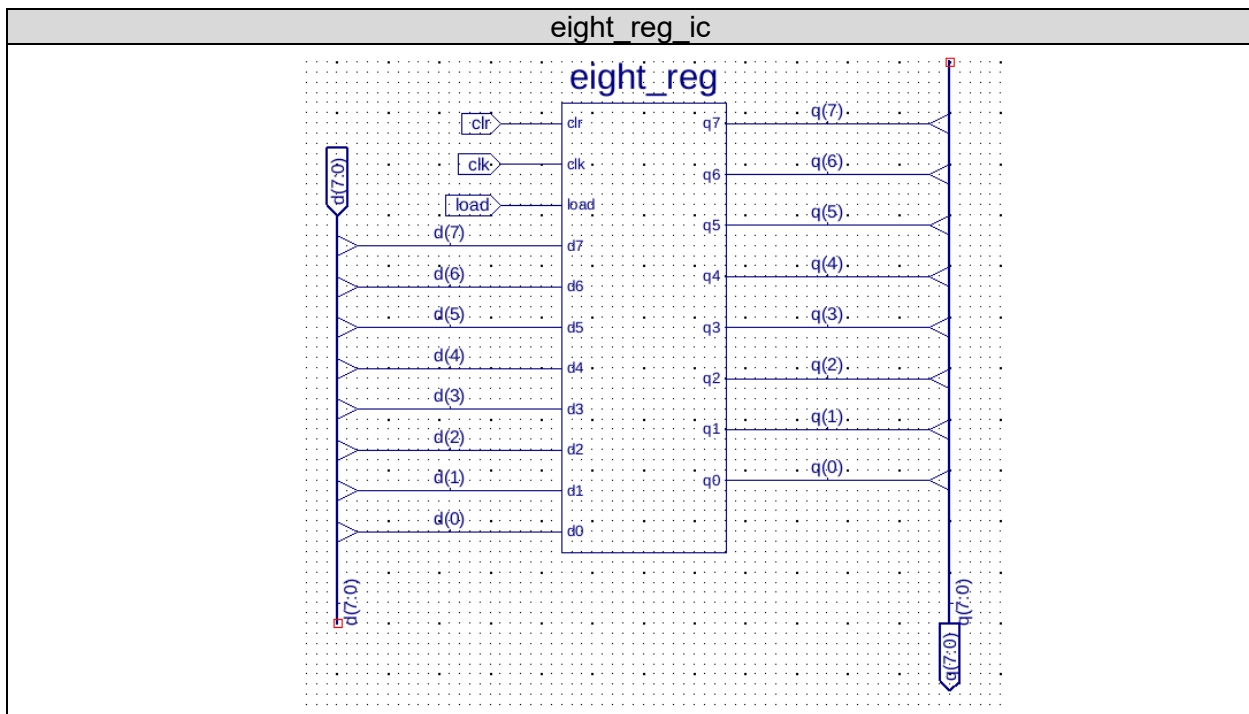
If the load signal of the clock is high, the 8-bit register must load all bits at the same time.

The asynchronous clear must reset the register to zero immediately and have priority over the clock and load signals.

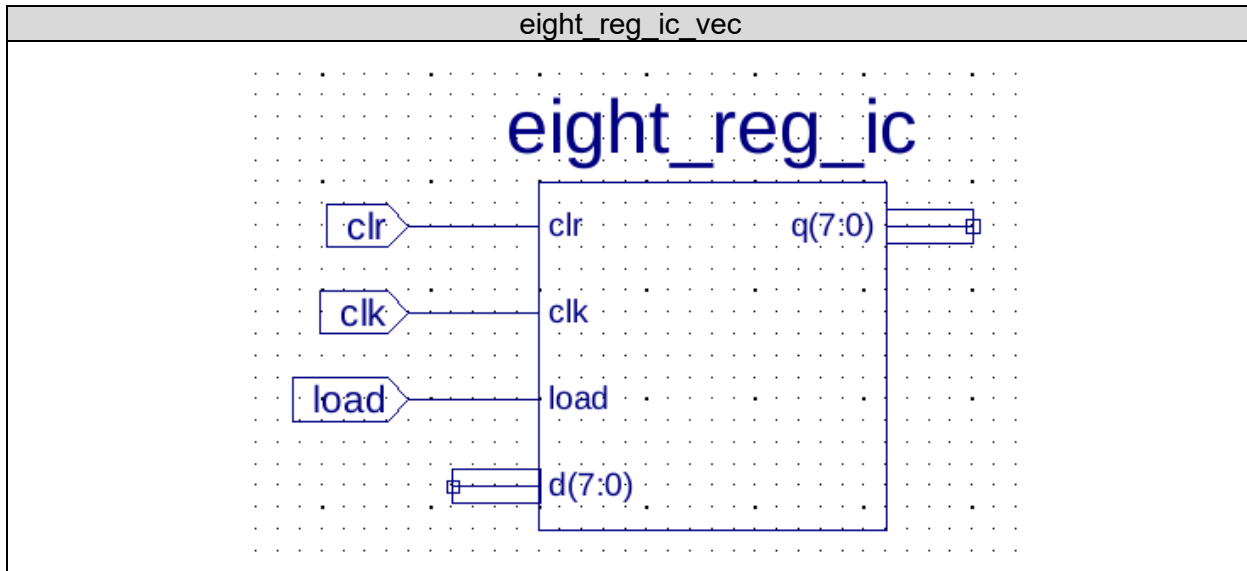
1. Eight-Bit Register



2. Eight-Bit Register IC



3. Eight-Bit Register IC (Vectorized)



4. Test Bench

Test Bench Code

```
// Verilog test fixture created from schematic
/home/ise/four_eight_reg/eight_reg_ic.sch - Sun Jan 18 11:33:00 2026

`timescale 1ns / 1ps

module eight_reg_ic_eight_reg_ic_sch_tb();

// Inputs
reg clr;
reg clk;
reg load;
reg [7:0] d;

// Output
wire [7:0] q;

// Bidirs

// Initialize Inputs
initial
begin
    clr = 1'b1;
    load = 1'b0;
    d = 8'b0000_0000;
end

initial
begin
    #5 clr = 1'b0;
end

endmodule
```

```

        load = 1'b1;

        #10 d = 8'b1110_0111;
    end

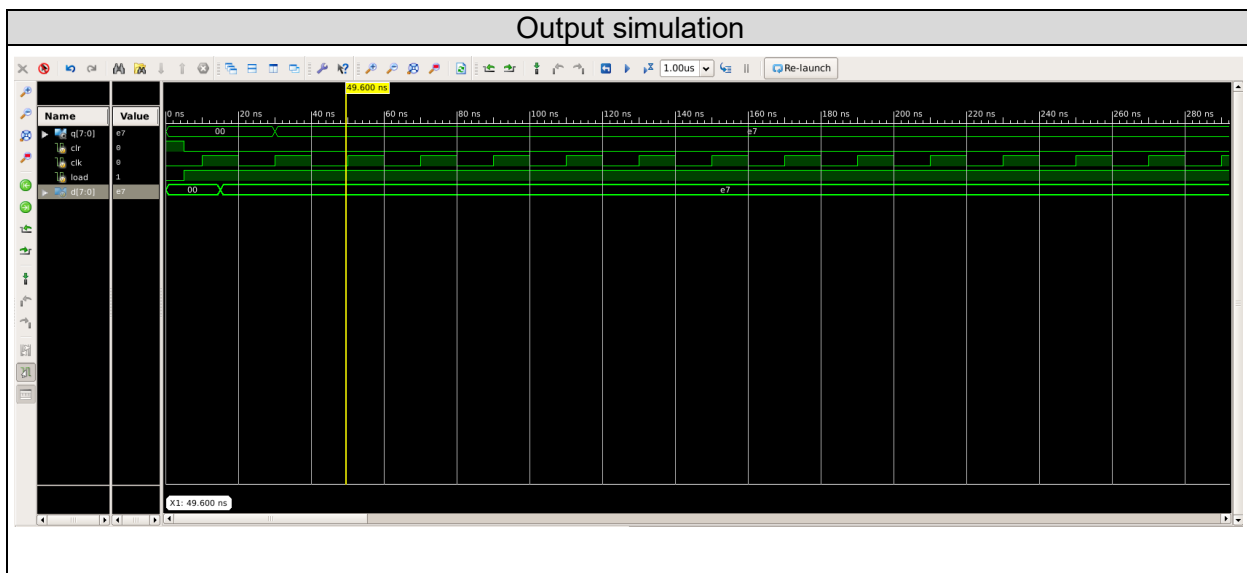
    initial
    begin
        clk = 1'b0;
        forever
            #10 clk = ~clk;
        end

    // Instantiate the UUT
    eight_reg_ic UUT (
        .clr(clr),
        .clk(clk),
        .load(load),
        .d(d),
        .q(q)
    );

endmodule

```

5. Output Simulation



References:

Digital Computer Electronics, 3rd Edition; Malvino, Albert Paul; Brown, Jerald A., McGraw-Hill, 1993

PCB Design Example:

https://www.youtube.com/watch?v=iEPjIELWarw&list=PLbUnUWOWn7M8QmODC1wdVhVPIP_u4FwBc&index=1

EasyEDA Dowload page

<https://easyeda.com/page/download>

SAP-1 Design

https://www.youtube.com/watch?v=oJvJXJb7uEU&list=PLbUnUWOWn7M8QmODC1wdVhVPIP_u4FwBc&index=2

https://www.youtube.com/watch?v=VzVTrh5uBjQ&list=PLbUnUWOWn7M8QmODC1wdVhVPIP_u4FwBc&index=3

https://www.youtube.com/watch?v=6xDo0wK8xAY&list=PLbUnUWOWn7M8QmODC1wdVhVPIP_u4FwBc&index=4

Build an 8-bit computer from scratch

<https://eater.net/8bit>

Simulator

<https://circuitverse.org/simulator>

SAP-1 Processor Architecture

<https://dangrie158.github.io/SAP-1/index.html>

<https://deeprajbhujel.blogspot.com/2015/12/sap-2-instruction-sets.html>

Digital Logic and Microprocessor Design with VHDL by Enoch O. Hwang